KAAN BOREKCI

R&D Engineer | Medical Devices | Robotics | AI for Healthcare

US Citizen | kaan@borekci.com
Portfolio | GitHub | LinkedIn

SUMMARY

R&D Engineer with 2+ years of experience in medical device development, applied machine learning, and fluid mechanics. Skilled in Python, embedded C/C++, PCB design, and Ansys Fluent. Leverage full-stack hardware and firmware, mechanics, and ML expertise to translate clinical requirements into deployable solutions.

EDUCATION

Northwestern University — Evanston, IL
Master of Science in Mechanical Engineering | GPA: 3.9/4.0 — Expected Jun 2026
Bachelor of Science in Biomedical Engineering | GPA: 3.5/4.0 — Jun 2025

SKILLS

Programming: Python, Embedded C/C++, MATLAB, Git, MMPose, OpenCV, PyTorch, SciPy, NumPy, Pandas
Tools: Ansys Fluent, Abaqus, EAGLE, Onshape, Solidworks, Raspberry Pi, Arduino, STM32, ESP32, PIC32, nRF
Languages: English, Turkish, German
Interests: Volleyball, Golf, Skiing, Tennis

WORK EXPERIENCE

Lurie Children's Hospital & Northwestern University — Chicago, IL
R&D Engineering Consultant | *Wearable Respiratory-Motion Platform for Pediatric CCHS Team* — Sep 2024 – Present

- Directed 4-person team in developing synchronized IMU and EtCO2 wearable platform, establishing foundation for closed-loop diaphragm pacer in pediatric CCHS
- Programmed dual-core RP2040 firmware using C, synchronizing 100 Hz IMU and 20 Hz CO2 streams over I2C/UART/SPI for clinical-grade SD logging
- Designed backup through-hole PCB in 1 week, preserving 15 weeks of project timeline after 2 failed SMD iterations and enabling IRB, provisional patent, and BMES submissions
- Collaborated with 4 clinicians to extract clinically relevant features from IMU and CO2 data in Python, validating device feasibility for future closed-loop pacing

Shirley Ryan Ability Lab, Center for Bionic Medicine — Chicago, IL
R&D Engineer | *Biomedical Machine Learning and Motion Analysis* — May 2024 – Present

- Collaborated with 4-person cross-functional team to translate General Movement Assessment criteria into 10+ quantitative pose-evaluation metrics for selecting clinically valid pose models
- Constructed 3D pseudo-ground-truth dataset across 17 infant patient videos via depth-camera and 2D keypoint fusion to validate downstream pose-lifting models
- Identified optimal pose lifter from 8 MMPose candidates by executing custom Python pipeline to evaluate across 375K+ frames and 17 keypoints of infant motion data
- Generated 50% more kinematic features across 300+ uncalibrated videos via validated pose lifter, targeting 12-month earlier motor-delay screening

International Institute for Nanotechnology — Evanston, IL
Nanotechnology Intern — Jun 2023 – Jul 2023

- Developed ImageJ workflow analyzing 80+ hours of video frames to identify optimal NaCl and temperature crystallization conditions
- Optimized DNA-coated nanoparticles to form well-faceted crystals 67% faster than 3-day cooling standard

Northwestern University | Innovation and New Ventures Office Evanston, IL
Technology Transfer Intern Jul 2022 – Aug 2022
- Developed queryable Excel database from Pitchbook, SBIR/STTR, and internal records, tracking $2.78B across 180+ Northwestern ventures over 12 years
- Collaborated with executive team to audit capital raised across 5 sectors and Seed-to-IPO stages, executing IP and commercialization strategy

Querrey Simpson Institute for Bioelectronics Evanston, IL
Bioelectronics Research Intern May 2020 – Sep 2020; Jul 2019 – Aug 2019
- Engineered synthetic skin models across 3 thicknesses and physiological flow rates, validating shunt flow detection device against patient temperature data
- Analyzed sensor data in Python, correlating temperature changes with thermal conductivity, to optimize device sensitivity for skin flap perfusion monitoring

PROJECT EXPERIENCE

NU | Physics-Informed Autoencoder for Pose-to-Dynamics Recovery March 2026 – June 2026
- Engineered physics-informed autoencoder in PyTorch with BiLSTM encoder and differentiable ODE decoder enforcing rigid-body dynamics, recovering 4 physical parameters from 2D pose video
- Validated model against data-driven baseline, improving projection geometry recovery R^2 from 0.81 to 0.92 and reducing natural frequency RMSE 15%

NU | CFD Aerodynamical Analysis of Aerosol in Lung Model Jan 2026 – Mar 2026
- Generated 3 structured meshes with y+ < 1 inflation layers for SST k-omega per ASME GCI methodology, achieving 7% convergence on deposition fraction
- Modeled Lagrangian particle transport in the upper airway using Ansys Fluent Discrete Phase Model with 500 streams and 50k parcels, reproducing 0.03%, 20.7%, and 100% Stokes deposition

NU | 6-DOF UR5 Robot Arm Motion Planning and Control Simulation Sep 2025 – Dec 2025
- Engineered Python pipeline converting end-effector SE(3) poses into 6-DOF joint trajectories via forward kinematics, inverse kinematics, and rigid-body dynamics
- Evaluated computed-torque controller across critically damped, underdamped, and torque-saturated regimes, validating physical feasibility under actuation limits

NU | Continuous Live Web Streaming Device Jan 2025 – Mar 2025
- Collaborated in 2-person team to deliver full-stack streaming prototype in 10 weeks with custom Eagle PCB, ESP32 networking, and 3D-printed enclosure
- Developed SAM4S firmware in C integrating camera + ESP32 via SPI/UART, enabling 320x240 live stream

PATENTS & PUBLICATIONS

- **Borekci, K.**, et al. (2025). "Wearable IMU-CO2 Device for Monitoring Activity-Dependent Respiratory Demand", U.S. Provisional Patent Application (Northwestern University, LUNU2025-213); filed Aug 2025. Status: Patent Pending.
- Krishnan, S. R., Arafa, H. M., Kwon, K., et al., **Börekçi, K.,** et al., & Rogers, J. A. (2020). Continuous, noninvasive wireless monitoring of flow of cerebrospinal fluid through shunts in patients with hydrocephalus. npj Digital Medicine, 3, 29. https://doi.org/10.1038/s41746-020-0239-1